Sequences & Series and Financial Mathematics — Vocabulary

数列与级数 · 金融数学 — 数学词汇表（英汉对照）

HSC Mathematics Advanced (MA12-4) • Words and phrases taken from the Sequences & Series practice papers and the Financial Mathematics practice papers (2023–2024 and 2025–2026 school trial questions)

How to use this list 使用说明: Each row gives the English term, its Chinese meaning, and a short explanation or an example sentence copied from the practice papers. Learn the words in Sections 1–4 first (they appear in almost every question), then the exam command words in Section 6. Part B covers the Financial Mathematics papers. 先掌握第 1–4 部分的核心词汇，再学习第 6 部分的考试指令用语；第二部分为金融数学词汇。

1. Basic words for sequences and series　一、数列与级数的基本词汇

English	中文	Meaning / example from the papers 含义 / 试卷例句
sequence	数列	An ordered list of numbers: 5, 9, 13, 17, … Each number is a *term*.
series	级数（数列各项的和）	The terms added together: 5 + 9 + 13 + 17 + … A series is the *sum* of a sequence.
term	项	One number in a sequence. “Find the 20th term of –7, –3, 1, 5, …”
first term (a)	首项	The first number in the sequence, written a or T_1. “The first term is –2.”
second term, third term …	第二项、第三项……	T_2, T_3, … “The third term is 8.”
***n*th term (general term), T_n**	第 n 项（通项）	A formula giving any term from its position n: $T_n = 58 - 7n$
general term / formula for the *n*th term	通项公式	“Write a simplified expression for the nth term of this sequence.”
last term (l)	末项	The final term of a finite sequence: 51, 44, 37, …, –47 (last term –47)
number of terms (n)	项数	How many terms there are. “How many terms are there in the sequence?”
position (of a term)	项的位置（序号）	The value of n: the 10th term has position $n = 10$.
subscript	下标	The small number/letter below the line in T_{10}, S_n, a_1.
consecutive terms	连续项 / 相邻项	Terms next to each other: “$x + 6$, $2x - 5$, $4x - 11$ are consecutive terms.”
successive terms	相继的项（前后两项）	Same meaning as consecutive. “the ratio between successive terms”
finite sequence / infinite sequence	有限数列 / 无限数列	Finite: has a last term. Infinite: continues forever (shown by “…”).
pattern	规律	“Assume the same pattern will be followed.”
increasing / decreasing sequence	递增 / 递减数列	Terms get bigger / smaller: 118, 105, 92, … is decreasing.
alternating (in sign)	正负交替	Terms change sign: $2x - 4x^2 + 8x^3 - \ldots$ (negative common ratio)
recursively defined sequence / recurrence relation	递推定义的数列 / 递推公式	Each term is defined from the previous one: $T_1 = 256$, $T_n = T_{n-1} \div 4$
previous term	前一项	T_{n-1} is the term before T_n.
sum of the first n terms, S_n	前 n 项和	$S_{10} = T_1 + T_2 + \ldots + T_{10}$. “The sum of the first 6 terms is 78.”
partial sum	部分和	Another name for S_n (the sum of only the first n terms).

2. Arithmetic sequences and series　二、等差数列与等差级数

English	中文	Meaning / example from the papers 含义 / 试卷例句
arithmetic sequence / arithmetic progression (AP)	等差数列	Each term is found by **adding** the same number: 5, 9, 13, 17, …
arithmetic series	等差级数（等差数列求和）	The sum of an arithmetic sequence: 2026 + 2011 + 1996 + … + 271

English	中文	Meaning / example from the papers 含义 / 试卷例句
common difference (d)	公差	The fixed amount added each time: $d = T_2 - T_1$. In 51, 44, 37, … $d = -7$.
constant difference	差为常数	"The difference between consecutive terms is constant" = the sequence is arithmetic.
arithmetic mean	等差中项	The middle term of three consecutive AP terms: $b = (a + c) \div 2$, i.e. $2b = a + c$.
$T_n = a + (n - 1)d$	等差数列通项公式	"a plus n minus one, times d." $T_{18} = 5 + 17 \times 4 = 73$
$S_n = \frac{n}{2}[2a + (n - 1)d]$	等差数列前 n 项和公式	Use when you know a, d and n.
$S_n = \frac{n}{2}(a + l)$	等差数列求和公式（首项加末项）	Use when you know the first and last terms. $S_{15} = 15⁄2 \times (51 + (-47)) = 30$
linear (in n)	关于 n 的一次式（线性）	$T_n = 6n - 7$ is linear: the coefficient of n is the common difference.

3. Geometric sequences and series 三、等比数列与等比级数

English	中文	Meaning / example from the papers 含义 / 试卷例句
geometric sequence / geometric progression (GP)	等比数列	Each term is found by **multiplying** by the same number: 18, 6, 2, …
geometric series	等比级数（等比数列求和）	The sum of a geometric sequence: 2 + 6 + 18 + … + 118 098
common ratio (r)	公比	The fixed multiplier: $r = T_2 \div T_1$. In 96, 72, 54, … $r = 3⁄4$.
constant ratio	比为常数	"Show that the terms form a GP" → show $T_2⁄T_1 = T_3⁄T_2$.
geometric mean	等比中项	The middle term of three consecutive GP terms: $b^2 = ac$. "x^2, $4x + 3$, 25 are in a GP."
$T_n = ar^{n-1}$	等比数列通项公式	"a times r to the power n minus one." $T_8 = 18 \times (1⁄3)^7$
$S_n = a(r^n - 1)⁄(r - 1)$	等比数列前 n 项和公式	Also written $a(1 - r^n)⁄(1 - r)$; valid for $r \neq 1$.
positive / negative common ratio	正公比 / 负公比	"The common ratio is negative" → terms alternate in sign, e.g. $r = -3⁄2$.
power / index / exponent	幂 / 指数	In 1.025^{28}, the index (exponent) is 28. "raise to the power of"
exponential growth / decay	指数增长 / 指数衰减	A GP with $r > 1$ grows; with $0 < r < 1$ it decays (e.g. 10 200 L of water leaking, 60% remaining).

4. Limiting sums (infinite geometric series) 四、极限和（无穷等比级数）

English	中文	Meaning / example from the papers 含义 / 试卷例句
limiting sum / sum to infinity, S_∞	极限和 / 无穷项和	The value the sum approaches when infinitely many terms are added: $S_\infty = a⁄(1 - r)$.
the limiting sum exists	极限和存在	Only when $\lvert r\rvert < 1$, i.e. $-1 < r < 1$. "Show that the limiting sum of this series exists."
converge / convergent	收敛 / 收敛的	The partial sums get closer and closer to a fixed number ($\lvert r\rvert < 1$).
diverge / divergent	发散 / 发散的	The sums do not approach a fixed number ($\lvert r\rvert \geq 1$): 1 + 1 + 1 + … has no limiting sum.
approaches / tends to	趋近于	"The total approaches the limiting sum \$8100." $r^n \to 0$ as $n \to \infty$.
infinity (∞)	无穷大	"as n tends to infinity"; S_∞ is read "S infinity".
absolute value / modulus, $\lvert r\rvert$	绝对值	$\lvert r\rvert < 1$ means the size of r is less than 1, ignoring the sign.
for what values of x …	当 x 取什么值时……	"For what values of x will this series have a limiting sum?" → solve $\lvert r\rvert < 1$ for x.

English	中文	Meaning / example from the papers 含义 / 试卷例句
interval	区间	"in the interval $-\pi/2 < \theta < \pi/2$" — the set of allowed values.

Symbols and notation 符号与记法

Symbol	读法 / 含义	Example
a	首项 first term	$T_1 = a$
d	公差 common difference	$d = T_n - T_{n-1}$
r	公比 common ratio	$r = T_n \div T_{n-1}$
n	项数 / 项的序号 number of terms / position	a positive integer 1, 2, 3, …
l	末项 last term	$l = T_n$
T_n (or a_n, t_n, u_n)	第 n 项 the nth term	T_{10} is read "T ten" or "the tenth term"
S_n	前 n 项和 sum of the first n terms	$T_n = S_n - S_{n-1}$
S_∞	极限和 limiting sum	$S_\infty = a/(1 - r)$, $\lvert r\rvert < 1$
…	省略号 "and so on"	shows the sequence continues
⇒	推出 "implies / therefore"	$4r^2 = 9 \Rightarrow r = \pm 3/2$
$\lvert r\rvert$	绝对值 absolute value of r	$\lvert -2/3\rvert = 2/3$
≈	约等于 approximately equal to	$S_{10} \approx 446.31$

5. Working and solving: general mathematics words 五、解题过程中的常用数学词汇

English	中文	Meaning / example from the papers 含义 / 试卷例句
formula	公式	"Use the formula for S_n."
substitute (into)	代入	Replace a letter by a number or expression: substitute $n = 20$ into T_n.
simplify / simplified expression	化简 / 最简表达式	Write in the shortest form: $51 + (n - 1)(-7) = 58 - 7n$.
simplest form	最简形式	"Find the limiting sum in simplest form." (e.g. $\cos^2\theta$, not $1/(1 + \tan^2\theta)$)
evaluate	求值	Work out the number: "Evaluate the limiting sum of $5/6 + 5/27 + \ldots$"
expression	表达式	A mathematical phrase without "=", e.g. $2n - 1$.
equation	方程	A statement with "=", e.g. $58 - 7n = -47$. Solve it to find n.
inequality	不等式	A statement with <, >, ≤ or ≥: $7n - 18 < 100$.
simultaneous equations	联立方程（方程组）	Two equations solved together: $a + 5d = 37$ and $a + 18d = 193$.
subtract / subtracting	相减	"Subtracting the equations gives $13d = 156$."
quadratic equation	二次方程	An equation with n^2: $n^2 + 7n - 408 = 0$.
factorise	因式分解	$(n + 24)(n - 17) = 0$
logarithm / natural logarithm (ln)	对数 / 自然对数	Used to solve for n in the index: $1.004^n > 11/6 \rightarrow n > \ln(11/6) \div \ln 1.004$.
ratio	比	$T_6 \div T_3 = r^3$
sum / difference / product / quotient	和 / 差 / 积 / 商	Results of +, −, ×, ÷.
whole number / integer / positive integer	整数 / 正整数	n must be a positive integer: from $n = 9$ or 5.5, choose $n = 9$.
coefficient	系数	In $6n - 7$ the coefficient of n is 6.
constant	常数	A fixed number, e.g. the common difference d.

English	中文	Meaning / example from the papers 含义 / 试卷例句
percentage / per cent (%)	百分比	“over 60% of the water remains”; a 25% increase means $r = 1.25$.
exact value / exact form	精确值 / 精确形式	Leave as a fraction, surd or log: $1024⁄3$, $(\sqrt{3}+1)⁄4$, ln 2 — not a rounded decimal.
correct to two decimal places	精确到小数点后两位	446.31 (two digits after the decimal point).
to the nearest dollar / cent / whole number	精确到元 / 分 / 整数	$158 (nearest dollar); $1001.76 (nearest cent).
minimum / maximum number of terms	最少 / 最多项数	“Find the minimum number of terms needed for the sum to be negative.”
the first term that is greater than …	第一个大于……的项	“Find the first term of $-12 - 8 - 4 + \ldots$ that is greater than 150.”
exceed / exceeds	超过	“the first night by which the total exceeds 1200”; “acidity exceeds 7”

6. Exam command words and question phrases 六、考试指令用语与题目常用表达

English	中文	Meaning / example from the papers 含义 / 试卷例句
Find	求	Work out the answer, showing working.
Calculate / Determine	计算 / 确定	Same as Find (usually a numerical answer).
Write down / State	写出 / 说明	No working needed: “State the common ratio.”
Show that …	证明…… / 说明……	The answer is given; you must write every step that leads to it.
Prove	证明	Give a complete logical argument.
Hence	由此（利用上一问的结果）	Use the previous part's result: “Hence find n.”
Hence, or otherwise	由此或用其他方法	You may use the previous part, or any other correct method.
Justify / Explain / Give a reason	说明理由 / 解释	“Justify whether this series has a limiting sum” → state $\lvert r\rvert < 1$ and why.
Explain why it does not exist	解释为什么不存在	e.g. “$\lvert r\rvert \geq 1$ so the series does not converge.”
in terms of	用……表示	“Find T_n in terms of n”; “in terms of a” means the answer contains the letter a.
Given that …	已知……	Information you may use without proving it.
Consider the series …	考虑级数……	Introduces the object of the question.
respectively	分别地	“A and B have terms a_i and b_i respectively” → $A \leftrightarrow a_i$, $B \leftrightarrow b_i$.
Which of the following …?	下列哪一个……？	Multiple-choice stem; choose A, B, C or D.
could NOT be said / is NOT true	不能说…… / 不正确的是……	Find the false statement.
Assume / Assuming the same pattern continues	假设规律不变	Treat the sequence as continuing with the same d or r.
Marks	分值	The number in the right margin shows how many marks the part is worth.

7. Context words from the application questions 七、应用题中的情境词汇

English	中文	Meaning / example from the papers 含义 / 试卷例句
deposit (money)	存入（存款）	“She deposits $100 on his first birthday.”
withdraw / withdrawal	取出 / 取款	“Jackson withdraws $2000 at the end of each month.”
donation / fundraising drive	捐款 / 募捐活动	“The weekly donations follow a geometric sequence.”
salary / wage / pay rise	薪水 / 工资 / 加薪	“His wage will increase by $6000 each year.”

English	中文	Meaning / example from the papers 含义 / 试卷例句
sales / total sales	销售量 / 总销售量	“After how many months will Sampson's total sales exceed Frontier's?”
profit / loss	盈利 / 亏损	Positive terms are profits, negative terms are losses.
leak / leaks out	漏出	“In the first hour 200 litres leak out.”
remain / remaining	剩余	“over 60% of the water remains in the tank”
adjacent	相邻的	“the difference between the lengths of adjacent strings is constant”
slat (of a blind)	百叶窗叶片	“A venetian blind consists of twenty slats.”
gap	间隙	“the gap between adjacent slats is 27 mm”
inscribed	内接的	“a sequence of squares is inscribed within a right-angled triangle”
similar triangles	相似三角形	Used to show $a_1 = 2a/3$.
hypotenuse	斜边	The longest side of a right-angled triangle.
area (of a square / rectangle)	面积	“the areas of the squares form a geometric sequence”
strip (rectangular strip)	矩形条	“rectangular strips of width ½ unit under the curve”
concentration / acidity (pH)	浓度 / 酸度（pH 值）	$H_n = 10^{-2}(½)^{n-1}$ is a GP; $A_n = 2 + (n-1)\log_{10}2$ is an AP.
night / month / quarter / week	夜 / 月 / 季度 / 周	Time periods used as the position n of a term.

Part B — Financial Mathematics Vocabulary

第二部分 — 金融数学词汇

Compound interest, annuities, loans and recurrence relations (MA12-2, MA12-4) • words taken from the Financial Mathematics practice papers

8. Interest, investments and accounts　八、利息、投资与账户

English	中文	Meaning / example from the papers 含义 / 试卷例句
interest	利息	Money paid for the use of money. "What will be the amount of interest earned?"
principal	本金	The original amount invested or borrowed, e.g. $6000 or a $720 000 loan.
interest rate / rate of interest	利率	"at an interest rate of 3% per annum"
per annum (p.a.)	每年（年利率）	Per year. "6% p.a." = 6% per year.
rate per period (*r*)	每期利率	Annual rate ÷ number of periods per year: 6% p.a. monthly → $r = 0.06 \div 12 = 0.005$.
simple interest	单利	Interest on the principal only (not used in these papers, but compare with compound interest).
compound interest	复利	Interest is added to the balance and then also earns interest: $A = P(1 + r)^n$.
compounded / compounding	复利计息	"8% p.a. compounded quarterly" = interest added 4 times a year at 2% each time.
compounded annually / half-yearly (six-monthly, biannually) / quarterly / monthly	按年 / 半年 / 季度 / 月复利	1 / 2 / 4 / 12 compounding periods per year.
compounding period / time period	计息周期 / 期	One interval between interest payments (a month, a quarter, a year …).
number of periods (*n*)	期数	25 years of monthly payments → $n = 300$.
invest / investment / investor	投资 / 投资者	"An investor wishes to accumulate $50 000 over 2 years."
account / savings account / investment account	账户 / 储蓄账户 / 投资账户	"deposits $900 at the end of each quarter into an account"
balance	余额	The amount in the account (or still owed) at a given time.
deposit (verb) / a deposit	存入 / 存款（注意：a deposit for a house = 首付）	"Heidi will deposit a fixed amount each month." (Also: a *deposit for a house* = the first payment when buying)
withdraw / withdrawal	取出 / 取款	"Jackson withdraws $2000 at the end of each month."
lump sum	一次性款项	A single payment: "Taylor invests a single lump sum today."
future value (FV)	终值	What an investment will be worth at the end: $FV = P(1 + r)^n$.
present value (PV)	现值	The amount needed now to reach a future value: $PV = FV(1 + r)^{-n}$.
accumulate (a sum)	累积（达到某金额）	"accumulate $50 000 over 2 years"
be worth / the value of the investment	价值	"Find how much his investment is worth after 10 years."
earn (interest) / interest earned	赚取（利息）/ 所得利息	"an account earning 6% p.a."; interest earned = final amount – total deposited.
grow / growth	增长	"The balance grows for 5 years at 5%."

9. Annuities and interest-factor tables　　九、年金与利息系数表

English	中文	Meaning / example from the papers　含义 / 试卷例句
annuity	年金	A series of equal payments made at regular intervals, earning compound interest.
contribution / contribute	供款 / 定期缴款	“Sam contributes $3000 at the end of each quarter.”
regular / equal payments (instalments)	定期等额付款（分期）	“Mark made equal instalments of $550 twice per year.”
at the end of each period	每期期末	Ordinary annuity: the usual case for the future-value table.
at the start / beginning of each period	每期期初	Each payment earns one extra period of interest: multiply by $(1 + r)$.
future value of an annuity	年金终值	$FV = M[(1 + r)^n - 1]/r$ — the total at the end of all payments.
present value of an annuity	年金现值	$PV = M[1 - (1 + r)^{-n}]/r$ — the lump sum that equals the payments.
future / present value interest factor	终值系数 / 现值系数	The value of an annuity of $1; multiply by the payment: FV = 900 × 34.48148.
table of interest factors	利息系数表	“Using the table provided, determine …” — read across the rate column and down the period row.
annuity of $1	每期 1 元的年金	The tables are for payments of $1 per period; scale up for other payments.
contribution per period for a future value of $1	终值为 1 元时的每期供款	The reciprocal of the FV factor: payment = target × factor.
superannuation (super)	养老金 / 退休公积金	“invests part of his superannuation into an annuity”
retire / retirement	退休	“Julio has just retired.”
savings / save (up)	储蓄 / 攒钱	“Liv plans to save $150 000 over the next 5 years.”
mature / maturity (of an investment)	到期	“when the first annuity matured” = when its payments finished.
remain invested	继续投资（不取出）	“the money remains invested for another 6 months”
run out of funds / empty the account	资金耗尽 / 账户清零	“How many withdrawals can he make before the account runs out of funds?”
eventually / overall	最终 / 总共	“How much interest did the bank pay overall?”

10. Loans and repayments　　十、贷款与还款

English	中文	Meaning / example from the papers　含义 / 试卷例句
loan	贷款	Money borrowed that must be repaid with interest.
borrow / borrower	借款 / 借款人	“Ella borrows $820 000 to purchase a house.”
lend / lender / bank / finance company	放贷 / 贷款方 / 银行 / 金融公司	“Joel borrowed $20 000 from a finance company.”
repay / repayment	偿还 / 还款	“equal monthly repayments of $6248 over 30 years”
monthly repayment (*M*)	月供（每月还款额）	“Find the monthly repayment M to the nearest cent.”
reducing-balance loan	余额递减贷款	Interest is charged each period on the amount still owing.
reducible interest	按余额计息	Same idea: “9% per annum reducible interest, calculated monthly”.
interest is charged / added	计收利息 / 加上利息	“Interest is charged before repayments are made every month.”
amount owing / balance owing / outstanding balance	欠款余额 / 未还余额	“Find the outstanding balance immediately after the 120th repayment.”
mortgage / home loan	房屋抵押贷款 / 住房贷款	“Naomi takes out a mortgage of $650 000.”
take out a loan	办理贷款	“A reducing-balance loan of $900 000 is taken out.”

English	中文	Meaning / example from the papers 含义 / 试卷例句
term (of a loan)	贷款期限	“throughout the term of the loan” — the total repayment time, e.g. 30 years.
pay off / fully repaid / cleared	还清	“the loan is paid off 25 years after the initial loan”
one-off payment	一次性付款	“Ella makes a one-off payment to reduce the balance to $720 000.”
increase the repayments	提高还款额	“the monthly repayments are increased by $500”
how many months faster	提前多少个月	Compare the new number of months with the original.
interest saved	节省的利息	Total paid with original repayments − total paid with higher repayments.
total interest paid	支付的利息总额	Total of all repayments − amount borrowed.
interest rate rises / falls	利率上升 / 下降	“the interest rate rises to 7.8% p.a.” — recalculate with the new r.
afford / can afford repayments of at most …	负担得起 / 最多能承担……	“He can afford repayments of at most $450 per month.”
minimum number of repayments	最少还款次数	The smallest n that makes the balance ≤ 0.
car loan / purchase	车贷 / 购买	“to purchase a car”
university fees / costs / expenses	大学学费 / 费用 / 开支	“to pay off his university fees”; “living expenses”

11. Recurrence relations and other question language 十一、递推关系及其他题目用语

English	中文	Meaning / example from the papers 含义 / 试卷例句
recurrence relation	递推关系	$A_n = A_{n-1}(1 + r) - M$ — each balance comes from the previous one.
A_n (amount after n periods)	第 n 期后的余额	“Let A_n be the amount owing after n monthly repayments.”
immediately after	紧接着……之后	“immediately after the interest has been paid” — order of interest and payment matters.
geometric series (in a loan formula)	（贷款公式中的）等比级数	$M(1 + 1.005 + 1.005^2 + \ldots)$ is a GP with ratio 1.005 — sum it with $a(r^n - 1)/(r - 1)$.
closed form / general formula	通项（封闭）公式	$A_n = P(1 + r)^n - M[(1 + r)^n - 1]/r$
set $A_n = 0$	令 $A_n = 0$（贷款还清）	The loan is fully repaid when the balance is zero; solve for M or n.
solve for n using logarithms	用对数解 n	$1.005^n = 5 \Rightarrow n = \ln 5 \div \ln 1.005 \approx 322.7 \rightarrow 323$ months.
round up (to the next whole period)	向上取整	A loan needing 322.7 payments takes 323 payments (the last one is smaller).
to the nearest cent / dollar	精确到分 / 元	$1518.94 (cent); $5157 (dollar).
Do NOT prove this	不必证明	You may use the given value without showing where it came from.
initial / initially	初始的 / 起初	“the initial interest rate”; “an initial deposit of $20 000”
subsequent	随后的	“each subsequent month”
inclusive	包括在内	“from Year 31 to Year 45 inclusive” = 15 years.
per cent increase / CPI (inflation)	百分比增长 / 消费者物价指数（通胀）	“increase by the CPI of 3% p.a.” → multiply by 1.03 each year.
wage / salary / pay rise	工资 / 薪水 / 加薪	“Ella gets a pay rise”
profit / loss	盈利 / 亏损	“How much profit or loss will he eventually make?”
expand a business	扩大业务	“takes a loan of $500 000 to expand their company”

Financial symbols and formulas 金融数学符号与公式

Symbol	读法 / 含义	Example
P	本金 principal	$A = P(1 + r)^n$
A or FV	终值 / 本利和 amount / future value	$FV = 6000(1.0025)^{60}$
PV	现值 present value	$PV = 30\,000 \div (1.03)^{60}$
r	每期利率 interest rate per period (as a decimal)	4.2% p.a. monthly $\rightarrow r = 0.0035$
n	期数 number of periods	8 years monthly $\rightarrow n = 96$
M	每期付款额 / 还款额 regular payment or repayment	"monthly repayment of $\$M$"
A_n	第 n 期后的余额 balance after n periods	$A_0 = P;\ A_n = A_{n-1}(1 + r) - M$
$(1 + r)^n$	复利增长因子 compound growth factor	$(1.005)^{360}$
\$ / c	元 / 分 dollars / cents	\$1001.76 = 1001 dollars 76 cents
%	百分比 per cent	8.4% = 0.084

Quick check 快速自测: Cover the Chinese column and say the meaning of each English term; then cover the English column and write the English word from the Chinese.
遮住中文栏，说出每个英文术语的含义；再遮住英文栏，根据中文写出英文单词。